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Systematic review and individual patient data analysis of pediatric head and neck squamous cell carcinoma: An analysis of 217 cases

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ABSTRACT

Introduction: Pediatric head and neck Squamous cell carcinoma (PHNSCC) is a rare disease. The optimum treatment and outcome remains poorly understood because of rarity.
Methods: We conducted an individual patient data analysis of PHNSCC. Two authors independently searched PubMed, google search, and Cochrane library for eligible studies using following search words: Pediatric Head and neck squamous cell carcinoma, Head and neck squamous cell carcinoma under age of 20, Head and neck squamous cell carcinoma in young, PHNSCC till June 1, 2016 published in English language.
Results: Total of 217 patients of PHNSCC were found in the literature. Median age among the cohort was 15 years (Range: 0–20 years) with a clear male preponderance. Oral cavity tumors were commonest 75 (70%) followed by laryngeal neoplasms 16(15%). Median disease free survival was 9 months (Range: 0–216 months). Median overall survival was 48 months (Range: 1–216 months). In univariate analysis treatment modality had significant impact on disease free survival (DFS). Whereas, patients treated with Surgery, Laryngeal primary had significantly better OS. Patients with associated fanconis anemia had significantly worse overall survival (OS).
Conclusion: PHNSCC is a rare disease with poorer outcome. Associated DNA defects leads to poorer OS. Patients treated with surgery alone or surgery followed by adjuvant radiation had better DFS and OS. Molecular profiling and personalized therapy may improve survival with limited toxicity.
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1. Introduction

Squamous cell carcinoma of head and neck (HNSCC) is considered disease of sixth or seventh decade of life. These tumors are most commonly associated with tobacco, alcohol consumption and ill-fitting dentures. HNSCC is rare entity in pediatric age group (≤20 years as defined by the American college of Pediatrics) [1–3]. Risk factors in Pediatric age group are unique including DNA repair defects like Fanconis anemia, Blooms syndrome, Xeroderma pigmenstosum, dyskeratosis congenita, connexin mutations, chronic graft-versus-host disease, epidermolysis bullosa, human papillomavirus infection(HPV), immunosuppressive states like organ transplant [4–10]. Because of rarity little agreement exists regarding optimum treatment options for these patients and often protocols for adult patients are advocated. But, younger age, anticipated morbidity and impact of associated risk factors make it difficult to formulate optimum treatment and outcome. SEER database analysis of 54 pediatric HNSCC (PHNSCC) reported superior outcomes in children compared to adults [11]. However, few other reports have reported dismal outcomes as well [12]. We intended to perform this systematic review and individual patient data analysis of published literature to assess the demographic profile, pattern of care and assess survival outcomes.

2. Materials and methods

2.1. Date source and search method

Two authors (BP and SM) independently searched PubMed, google search, and Cochrane library for eligible studies using following search words: pediatric Head and neck squamous cell carcinoma, Head and neck squamous cell carcinoma under age of

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20, Head and neck squamous cell carcinoma in young, PHNSCC till June 1, 2016 published in English language. The details of search strategy in PubMed have been mentioned in supplementary digital content. References from the primary search result also were manually searched for potentially eligible studies.

2.2. Study selection

We included only published articles describing HNSCC and treatment ≤20 years of age, fully published case reports or case series. Articles on pathology and genetics alone without report on treatment and clinical outcomes were excluded from the analysis. Two independent authors (BP) and(SM) selected the eligible studies. Any disagreement between two authors was solved by discussing with a third author (GKR). Study selection has been depicted in Fig. 1 PRISMA flow diagram.

2.3. Data collection and data items

Two authors independently (BP, SM) extracted all data from the eligible trials. The following data were collected from each of the study: name of the first author, year of publication, sex, age, site of disease, TNM staging, and treatment received, associated disease, disease free survival (DFS), Overall survival (OS). Initially, all data were tabulated in Microsoft Excel spread sheet. We neither asked the author(s) nor contacted the publishers for any unpublished data. Pooled statistical analyses were performed by ADU.

2.4. Statistical analysis

Data was analysed and categorical variables were summarized by frequency (%) and quantitative variables were summarized by median and range. Survival outcomes were calculated from the time of diagnosis. DFS was calculated from the time of treatment to local progression. Univariate analysis (log rank test) was used to find impact of different prognostic variables on DFS and OS, and p value of ≤0.05 was taken as significant. Kaplan Meier method was used for survival analysis. STATA 12 software was used for all statistical analysis.

3. Results

3.1. Patient characteristics

Total of 217 patients of PHNSCC were found in the literature. Individual patient data was available for 107 patients from 54 publications and statistical analysis was performed on 107 patients. Median age of the cohort was 15 years (Range: 0–20 years).

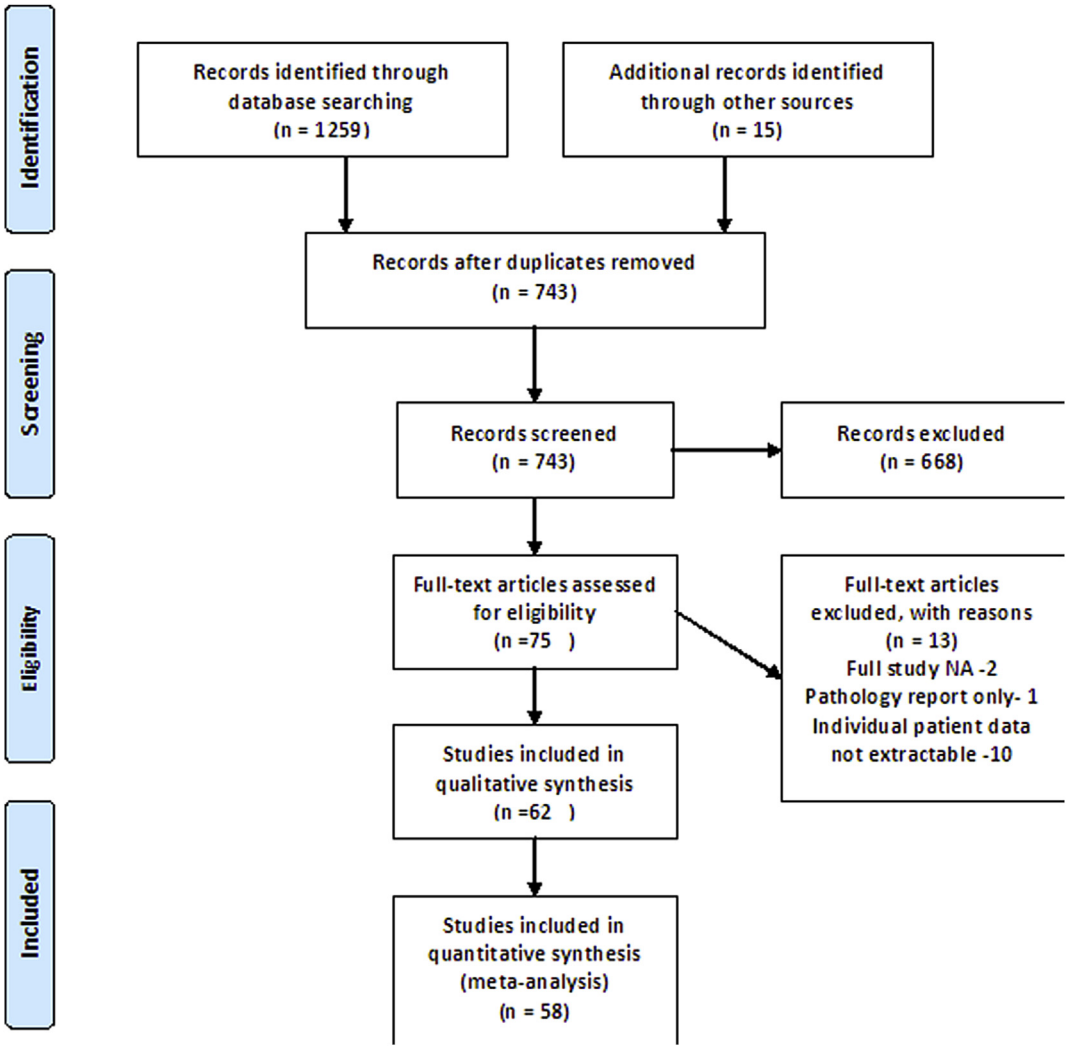

Fig. 1. Depicts PRISMA flow diagram for selection of publications.

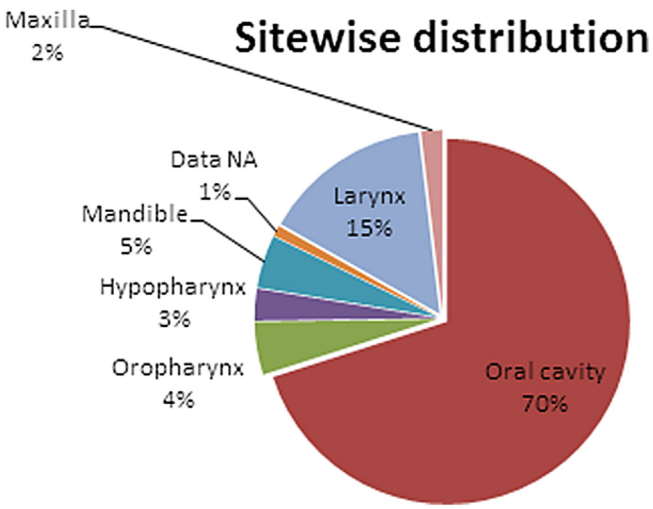

Fig. 2. Depicts site wise distribution of incidence of PHNSCC.

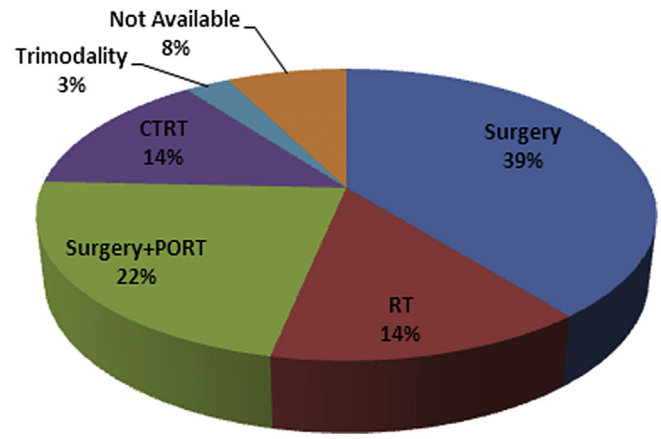

Fig. 4. Depicts use of different treatment modality for the treatment of PHNSCC.

22(20.6%) patients were below 10 years and 85(79.4%) were between 11 and 20 years of age. Sex ratio was skewed in favor of males with ratio of 1.7:1 (Male: 66, female: 38). Oral cavity tumors were commonest 75 (70%) followed by laryngeal neoplasms 16(15%) [Fig. 2]. Among oral cavity tumors, tongue-56(74.67%) was involved most commonly followed by gingiva 10(13.3%), retromolar trigone1(1.3%), Upper lip 1(1.3%), Lower lip 1(1.3%), buccal mucosa 4(5.3%), hard palate 1(1.3%), floor of mouth 1(1.3%) [Fig. 3]. 11(10.2%) patients had DNA repair defect, fanconis anemia and 9(8.4%) patients had other risk factors like smoking(3), tobacco chewing(1), HPV infection [1], xerodera pigmenstosa(2),retinoblastoma survior(1) and sharp teeth(1).

3.2. Pattern of care

Surgery alone was used in 42(39.3%), radiotherapy alone in 15(14%), surgery followed by adjuvant radiotherapy in 24(22.4%), concurrent radiotherapy and chemotherapy in 15(14%), trimodality therapy 3(2.1%). Treatment data was not available for 8 (7.5%) [Table 2]. Fig. 4 depicts the various treatment modalities used.

Out of 15 patients treated with radiation only external beam radiotherapy was used in 11 patients and brachytherapy was used in 4 patients. Interstitial brachytherapy with radon seeds and external beam was used in 2 patients, 10 g of potassium Iodide was used in 1 patient, and interstitial brachytherapy with neck dissection was used in 1 patient. Surgery with or without radiotherapy was used in 66 patients. Out of these 66 patients surgery for the primary site alone was done in 31 patients and surgery with neck dissection was done in 20 patients. Among 20 neck dissections 8 were selective neck dissections, 9 were modified neck dissection, 1 radical neck dissection and in 2 patient's neck dissection type were not specified. Brachytherapy for primary with neck dissection for nodes was used in 1, brachytherapy followed by wide local excision and neck dissection was done in 2 patients. Neoadjuvant chemotherapy followed by wide local excision and modified neck dissection followed by post-operative radiotherapy was used in 1 patient. Surgery followed by post-operative radiotherapy was used in 11 patients. Neoadjuvant chemotherapy was used in 4 patients among which 1 patient had docetaxel, cisplatin and 5 Fluorouracil and 1 patient received paclitaxel and carboplatin and data was NA in 2 patients. Brachytherapy with chemotherapy was used in 1 patient and WLE with adjuvant chemo-radiation was used in 1 patient. Palliative chemotherapy alone was used in 6 patients with

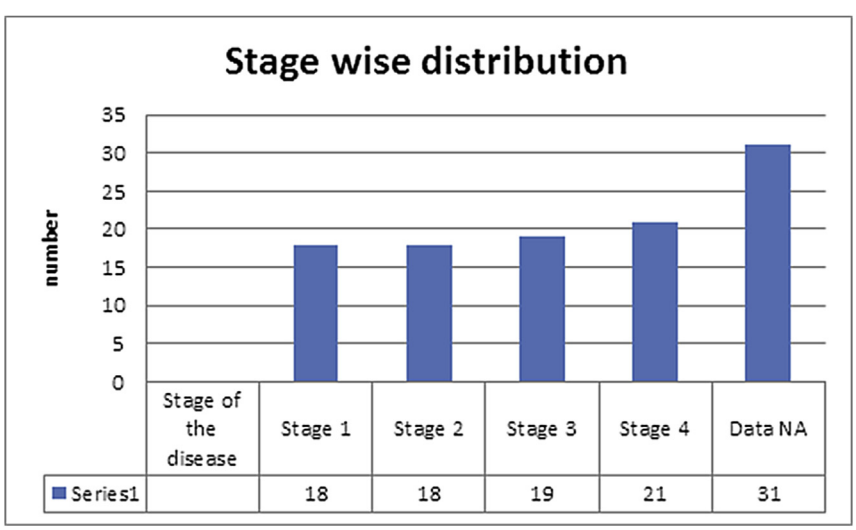

Fig. 3. Depicts Stage distribution of PHNSCC.

Table 1
Univariate analysis of factors affecting overall survival (OS) and disease free survival (DFS).

Factor	DFS			OS		
	2year DFS	95% CI	p value	2year OS	95% CI	p value
1. Year of publication						
≤2000	65.75%	0.48–0.79	0.16	67.41%	0.50–0.79	**0.025**
>2000	81.31%	0.60–0.92		44.62%	0.25–0.62	
2. Age in years						
≤10	75.75%	0.60–0.86	0.73	72.73%	0.42–0.88	0.456
>10	66.67%	0.05–0.95		54.43%	0.39–0.67	
3. Sex						
Male	74.64%	0.58–0.85	0.49	64.06%	0.46–0.77	0.18
Female	67.71%	0.45–0.82		47.24%	0.27–0.64	
3. Site of disease						
Oral cavity	75.75%	0.60–0.86	0.13	50.77%	0.34–0.65	**0.018**
Oropharynx	66.67%	0.05–0.95		66.67%	0.05–0.94	
Hypopharynx	Not reached			Not reached		
Mandible	100%			Not reached		
Data NA	–			–		
Larynx	58.3%	0.27–0.80		75%	0.40–0.91	
Maxilla	100%				Not reached	
4. Oral cavity primary	75.75%	0.59–0.86	0.31	50.77%	0.34–0.65	0.316
Non oral cavity primary	65.33%	0.42–0.80		69.57%	0.46–0.84	
5. Stage of the disease						
Stage 1	86.67%	0.56–0.96	0.12	65%	0.28–0.86	**0.009**
Stage 2	78.57%	0.47–0.93		81.82%	0.44–0.95	
Stage 3	40.80%	0.11–0.69		45.45%	0.16–0.70	
Stage 4	76.15%	0.43–0.92		59.53%	0.27–0.80	
6. Treatment modality						
Surgery alone	87.27%	0.70 0.95	**0.0013**	66.17%	0.45–0.80	**0.004**
Radiotherapy alone	28.57%	0.01–0.69		22.22%	0.03–0.51	
Surgery and radiotherapy	67.19%	0.38–0.85		68.57%	0.34–0.87	
Radiotherapy and chemotherapy	46.02%	0.13 0.74		37.50%	0.09–0.65	
Trimodality therapy	100%			Not reached		
Data NA	–					
7. Associated diseases						
Fanconi anemia and BMT	Not reached		0.71	0	–	**0.0001**
Others	70.04%	0.56–0.80		64.67%	0.49–0.76	
None	83.33%	0.27–0.97		100%	–	

P values that were significant are given in bold.

Hydroxyurea-1, Methotrexate-1, Methotrexate with cyclophosphamide-1, Retinoic acid and 5 Flououracil-1, data not available-2 patients.

3.3. Survival outcome

Median disease free survival was 9 months (Range: 0–216 months). 2 and 5 year DFS for oral cavity and non-oral cavity primary was 75.7%, 75.7% and 65.3%, 65.3% respectively. However, the difference was not statistically significant ($p = 0.31$). 5 year DFS was 87.2% for patients treated with surgery only and 67.1% for patients treated with surgery followed by adjuvant radiation. Whereas 5year DFS was 28.5%, for patients treated with radiation alone $p = 0.0013$. 5 year DFS for stage 1, 2, 3 were 86.6%, 78.5% and 40.8%

Table 2
Depicts stage wise distribution of different primary subsites and the treatment modalities used according to subsite and stage [RT-Radiotherapy, CT-Chemotherapy].

	Oral Cavity	Oropharynx	Hypopharynx	Mandible	Larynx	Maxilla
Surgery	Stage I-15	1	0	0	0	0
	Stage II-4	1	0	2	2	1
	Stage III-0	0	0	0	0	1
	Stage IV-2	0	0	0	0	0
Radiotherapy	Stage I-0	0	0	0	1	0
	Stage II-0	0	0	0	1	0
	Stage III-3	2	0	0	0	0
	Stage IV-3	0	0	0	0	0
Surgery + RT	Stage I-1	0	0	0	0	0
	Stage II-5	0	0	0	0	0
	Stage III-5	0	0	0	0	0
	Stage IV-8	0	0	0	0	0
CT + RT	Stage I-0	0	0	0	0	0
	Stage II-0	0	0	0	0	0
	Stage III-3	1	1	0	0	0
	Stage IV-4	0	2	0	0	0
Tri-modality	Stage I-0	0	0	0	0	0
	Stage II-0	0	0	0	0	0
	Stage III-1	0	0	0	0	0
	Stage IV-0	0	0	1	1	0
NA	21	0	0	2	11	0

respectively; p = 0.11. 2 and 5 year DFS for the entire cohort was 72.3% [95% CI: 0.59–0.81)] and 72.3% [95% CI: 0.59–0.81)]. Age, time of publication, sex, associated syndromes had no significant impact on DFS.

Median overall survival was 48 months (Range: 1–216 months). 2 and 5 year OS was 57.5% [95%CI: 0.44–0.68] and 44.8% [95% CI: 0.28–0.59] respectively. 2 year OS for patients treated with surgery alone, or surgery followed by adjuvant radiation, radiotherapy alone was 49.6%, 68.6% and 22.2% respectively. 5 year OS was 72.7% for patients ≤10 years of age compared to 54.4% for >10 years of age, but p value was not significant. Early stage patients had a significantly better survival compared to patients in advanced stage of disease (p = 0.0097). Patients treated with Surgery alone and surgery followed by adjuvant radiation had 5 year OS of 66.2% and

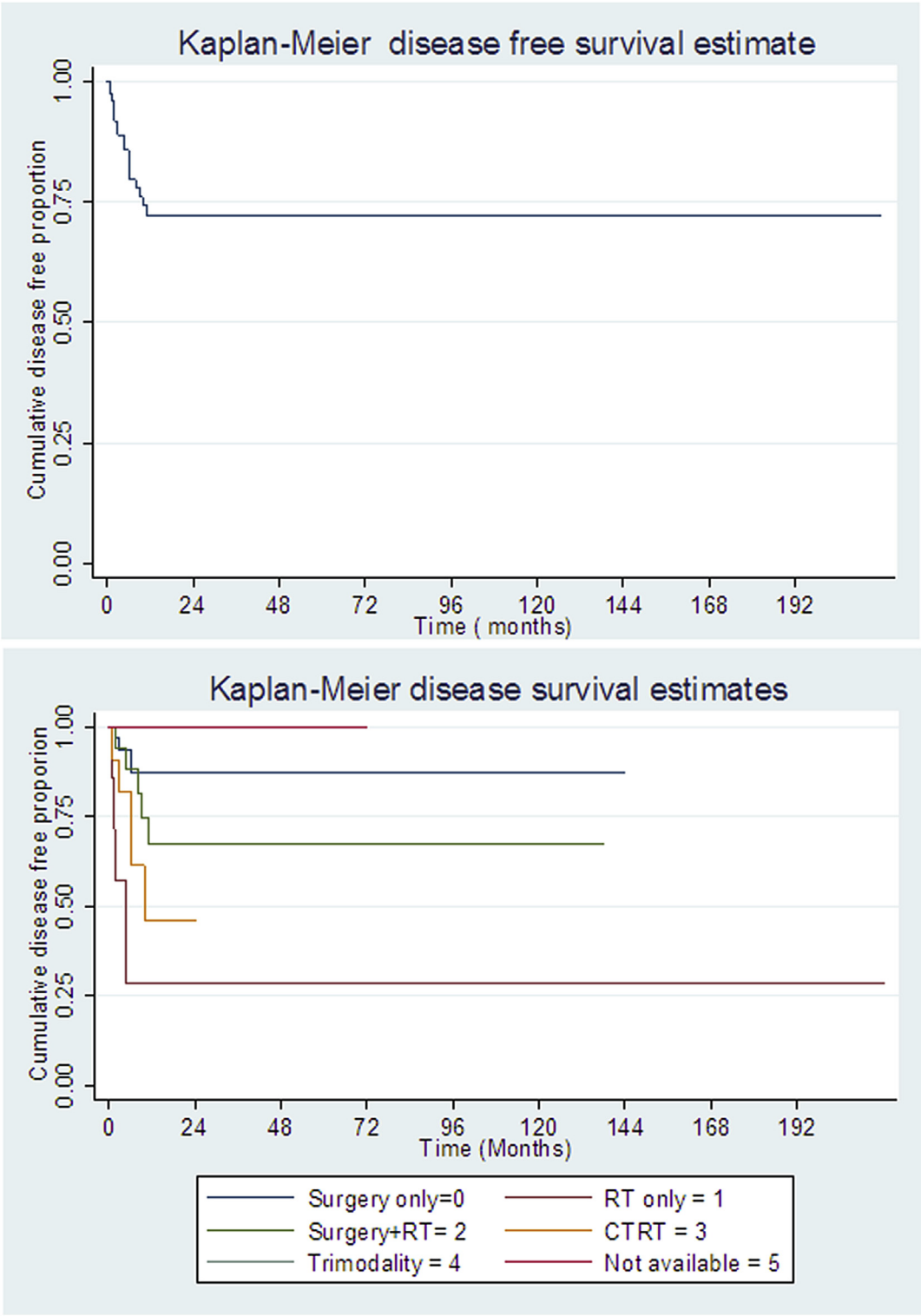

Fig. 5. Kaplan Meier survival curves of DFS and factors with impact on DFS.

68.6% respectively. The impact of treatment modality was significantly favouring surgery compared to radiotherapy alone and radio-chemotherapy, p = 0.004. 2 year OS was significantly worse for patients with associated Fanconis anemia, (p = 0.00001). Patients treated earlier than year 2000 had a significantly better survival compared to those treated after 2000 (p = 0.025). However, there was no impact of sex, oral cavity vs. non-oral cavity tumor on OS. The impact of different prognostic variables on DFS and OS has been tabulated in Table 1. Fig. 5 depicts Kaplan Meier curves for DFS of the entire cohort and impact of variables on DFS. Fig. 6 depicts Kaplan Meier curves for OS of the entire cohort and impact of variables on OS.

3.4. Pattern of failure

Recurrence was documented in 25(23.4%) patients. Local recurrences was most frequent in 17(68%), followed by local with simultaneous systemic failure in 2(8%), systemic alone (lung metastasis) in 6 (24%) patients. Recurrence was found in 11.1%, 16.6% and 33.3% patients of Stage I, stage II and Stage III respectively. 10% patients treated with surgery alone experienced recurrence, whereas 22.73% treated with surgery and adjuvant radiation had recurrence. 36.3% patients treated with radiation alone and 38.5% patients treated with chemo-radiotherapy had recurrence.

4. Discussion

The present analysis is aimed to look into the demography, pattern of care and survival outcome of pediatric patients with HNSCC. PHNSCC is a rarity as shown by only 217 cases over a century of literature search. SEER database analysis by Morris et al. reported female predisposition (37.0% vs. 31.7%, p = 0.04 [13]. But in the present study 66(63.46%) patients were male and 38 (36.54%) were female with a male preponderance. In contrast to adult HNSCC, nearly 81.34% patient had no documented risk factors. Risk factors were unique and ranged from DNA repair defects, the most common followed by fanconis anemia [14]. Though uncommon, early onset of tobacco chewing can be risk factor for PHNSCC as well, particularly in the developing countries [15]. 79.4% patients belong to the age group between 10 and 20 years compared to first decade of life. This finding further supports the hypothesis of delayed development of malignancy in patients with DNA repair defects. In an earlier report, out of 30 PHNSCC patients with systemic disease 21 had FA,4 had XP and 26 patients were in the age group of 10–20 years [16]. Oral cavity tumors were most common seen in 70%, with tongue being the most common affecting 64 children followed by gingiva. Tongue and gingiva has very high niche of stem cells which may be responsible for increased carcinogenesis from these subsites [17].PHNSCC have a tendency to present in advanced stage with delay in diagnosis due to lack of clinical suspicion and consideration of benign pathologies as differentials. Surgery was the most common treatment modality used in 40%.Surgery alone has been most commonly used since most of the patients had oral cavity tumor and most of them were in early stage of presentation. 2 year DFS and OS were best in surgery arm since 26(61%) of patients in surgery alone had stage 1 and stage 2 disease. Similar results were reported in the SEER analysis by Morris et al. as well [13]. In univariate analysis patients treated with surgery had significantly better DFS, OS compared to patients treated with radiation alone or surgery followed by adjuvant radiation. Outcomes in radiotherapy alone were dismal since intent in most cases was palliative only. 14% patients received concurrent chemo-radiation and 3% received tri-modality therapy. Brachytherapy has been used sparingly with no clear evidence to advocate usage in pediatric patients. 5 year DFS for stage 1, 2, 3 were 86.6%, 78.5% and 40.8% respectively. But, DFS for patients with stage 4 was 76.15% [95%CI (0.43–0.92)] similar to stage 2 which may be due to selective reporting and publication bias. These stage wise results, though not for separate subset of disease, looks no different from the adult counterparts. We intended to look into possible

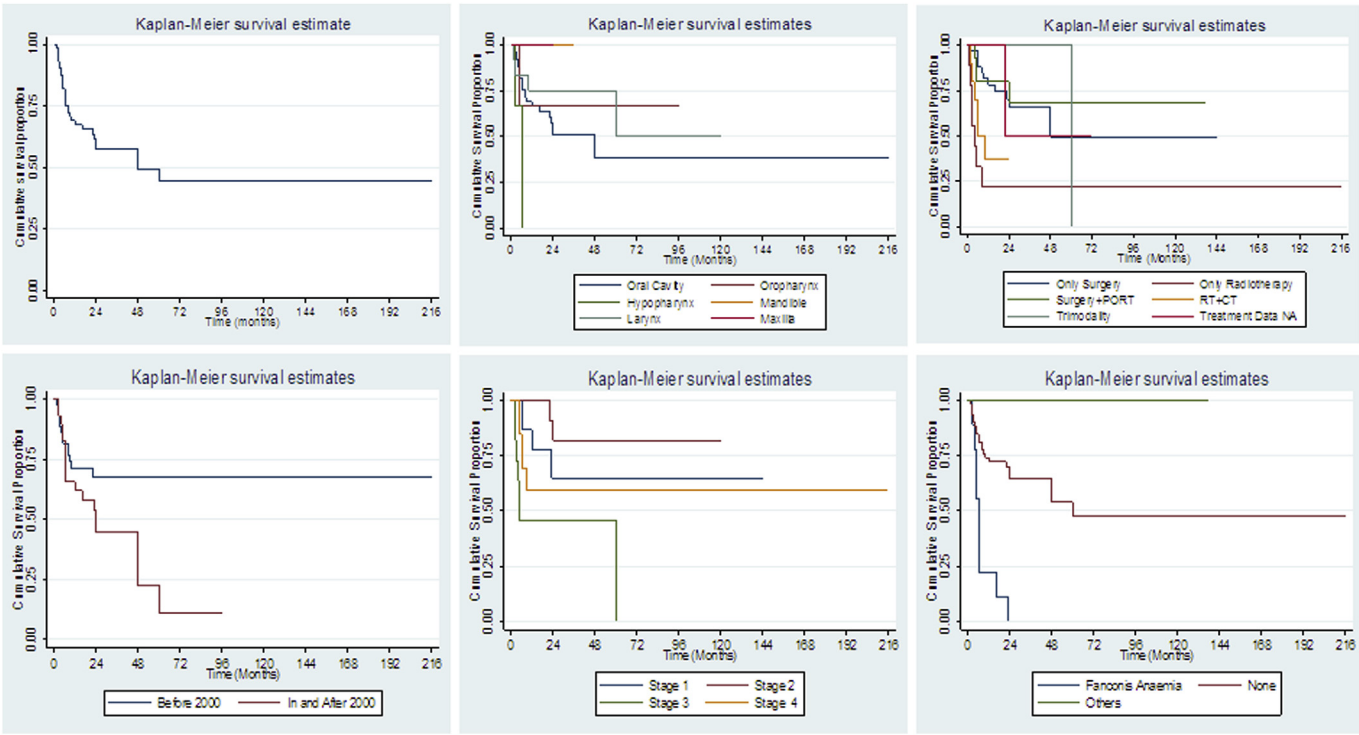

Fig. 6. Kaplan Meier survival curves of OS and factors with impact on OS.

prognostic factors. However, other than treatment modality none of the prognostic factors showed a significant impact on DFS. In the present analysis, patients with Fanconis anemia had significantly worse OS compared to patients with no known risk factors. Patients below 10 years of age, year of publication <2000, stage of disease, associated syndromes, and type of treatment modality employed had a significant impact on OS on univariate analysis.

This is the first systematic review of PHNSCC that has attempted to prognosticate patients and assess survival outcomes. The limitations of this systematic review are varied heterogeneity among studies reported, underreporting of data in various articles, wide variation in treatment modalities across the century and absence of toxicity reporting. In addition, these reports followed contemporary staging which makes it difficult to interpret and pool the data. Point should be made that enthusiasm of reporting such rare tumors leave behind the opportunity to report all important parameters. However, being a rare disease it is not possible to have large data and well described treatment options. This analysis will provide care givers an insight and a real world scenario to formulate optimum treatment option for PHNSCC patient and accumulate data for future analysis.

5. Conclusion

PHNSCC is extremely rare entity with distinct challenges to the physician and care givers. The treatment principles employed across institutes are similar to adults with outcomes comparable to adult counterparts. Surgery was the primary modality with best outcomes, as mostly early stage patients are offered such treatment. PHNSCC with systematic association has poor outcomes with dismal prognosis. Further molecular analysis of tissue specimens may throw some light on the exact etio-pathogenesis of this rare subset.

Disclosures

The authors have nothing to disclose.

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Compliance with ethical standards'

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* Research involving human participants and/or animals: Not applicable.
* Informed consent: Not applicable.

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